



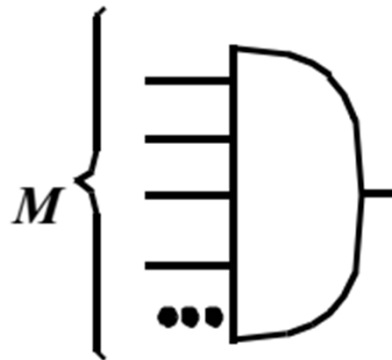
Heaven's Light is Our Guide

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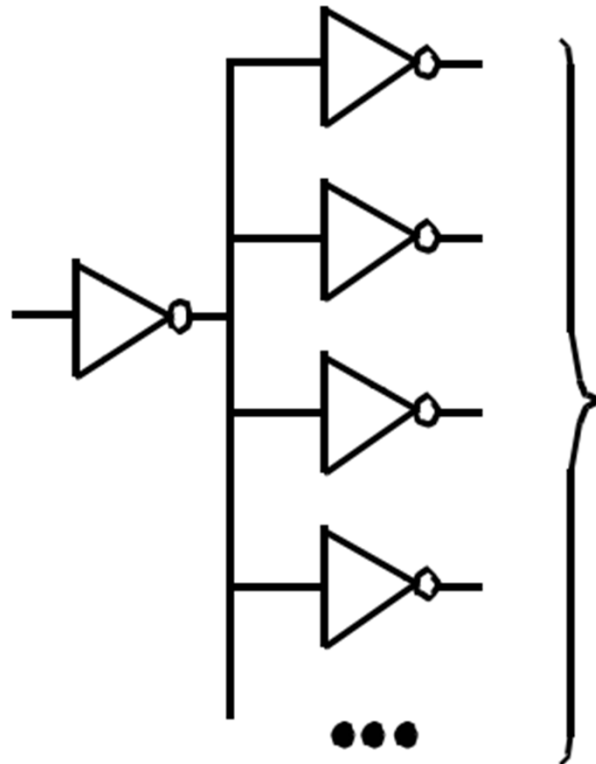
Course Code: CSE 2203
Course Title: Digital Techniques

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Fan in: Fan in is the **maximum number of inputs** connected to the gate without **any degradation** in the voltage level.



Fan-out: Fan out specifies the **maximum number of standard loads** that the output of the gate can drive without **impairment of its normal operation.**



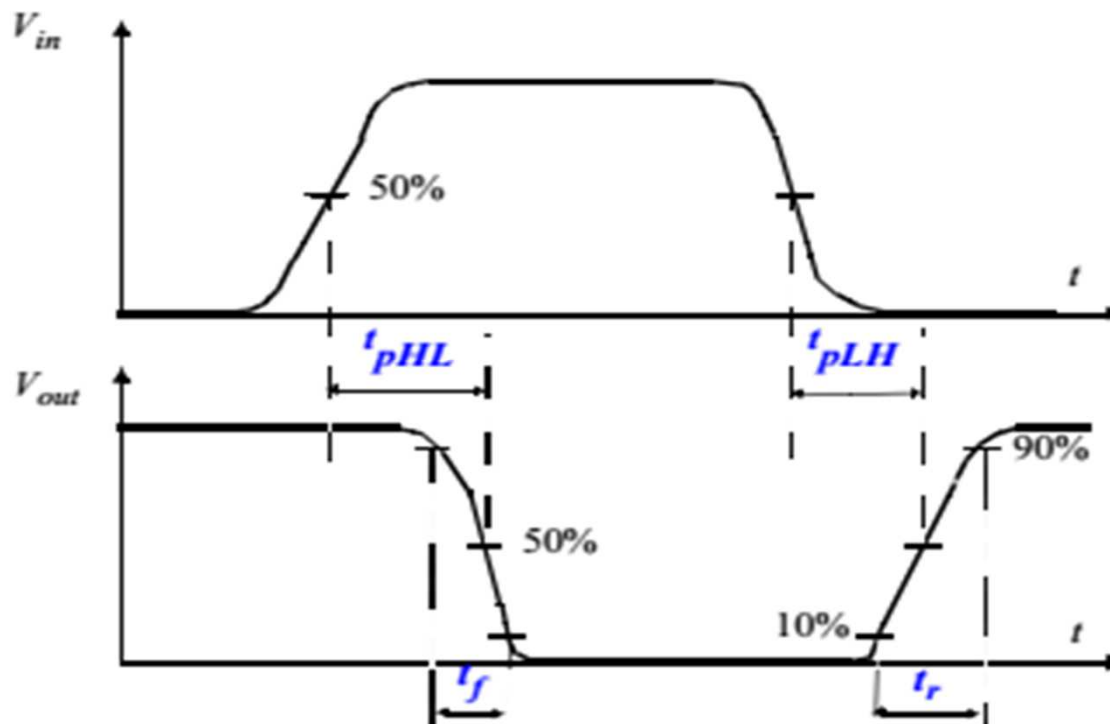
Power dissipation: Power dissipation is measure of power consumed by the gate when fully driven by all its inputs.

- Every IC requires a certain amount of electrical power to operate.
- Supplied by one or more power-supply voltages connected at V_{CC} (TTL) or V_{DD} (MOS devices).

Power dissipation: Power dissipation is measure of power consumed by the gate when fully driven by all its inputs.

- For many ICs, current drawn from the supply varies depending on logic states of the circuits on the chip.
- The amount of power an IC requires is determined by the current, I_{CC} (or I_{DD}) it draws from the supply.
- Actual power is the product of current and voltage.
- $P = I_{CC} \times V_{CC}$ or $I_{DD} \times V_{DD}$.

Propagation delay: Propagation delay is the **average transition delay time** for the signal to propagate from **input to output** when the signals change in value. It is expressed in ns.



Noise immunity refers to the circuit's ability to **tolerate noise** without **changes in output voltage**.

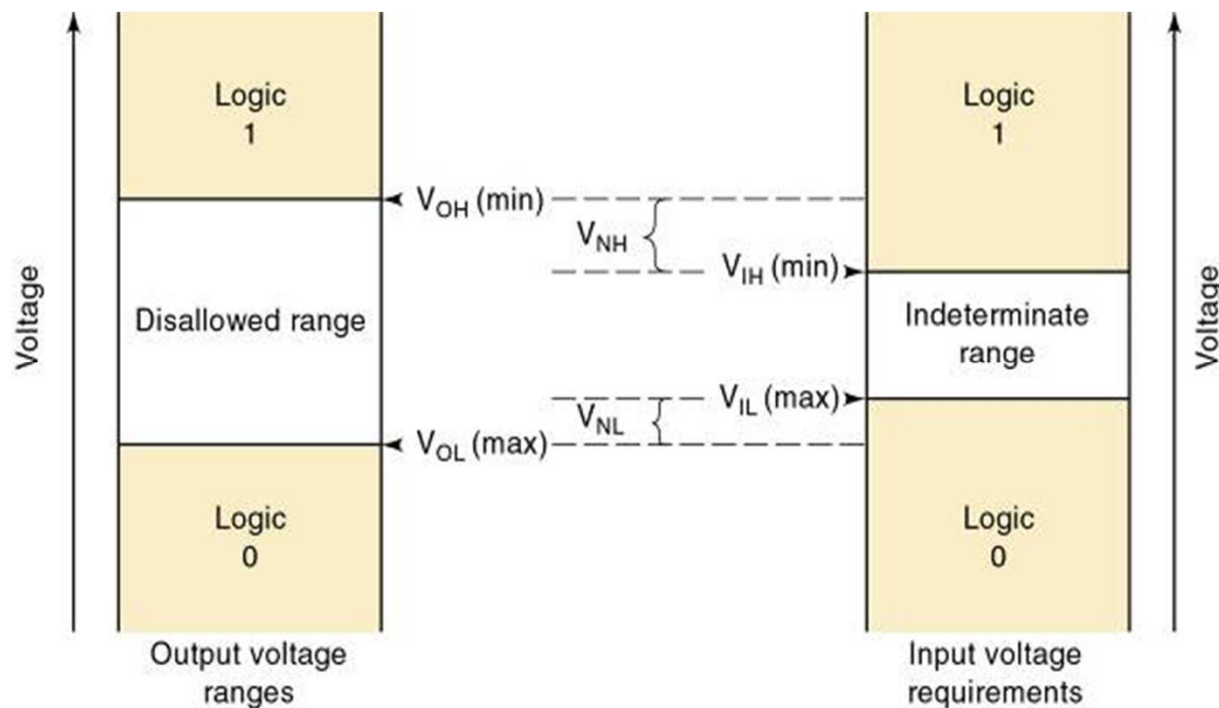
A quantitative measure is called **noise margin**.

High-state noise margin:

$$\begin{aligned} V_{NH} &= V_{OH}(\min) - V_{IH}(\min) \\ &= 2.4 \text{ V} - 2.0 \text{ V} = 0.4 \text{ V} \end{aligned}$$

Low-state noise margin:

$$V_{NL} = V_{IL}(\max) - V_{OL}(\max)$$



Noise margin:

- It is the **maximum noise voltage** added to an input signal of a digital circuit that does not cause an **undesirable change** in the circuit output.
- It is expressed in volts.

Thank You



Next topic